Assignment 1

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Econometrics I
NYU Stern

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Instructions. This assignment covers Lectures 1–2 only. You may use `R` or Python. Turn in (1) a PDF presenting your results and showing your work, and (2) your code. Derivations should use only the operator rules from Lecture 1 (linearity of $\mathbb{E}[\cdot]$, the variance/covariance rules, the law of iterated expectations, and the law of total variance) — if you find yourself writing an integral, you are working too hard.

1 The algebra of expectations

Let X and Y be random variables and a, b constants.

(a) Starting from the definition of expectation for a discrete random variable, show that $\mathbb{E}[a + bX] = a + b\,\mathbb{E}[X]$.

(b) Using only linearity (and the definition $Var(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$), show that $Var(a + bX) = b^2\, Var(X)$.

(c) Show that $Var(X + Y) = Var(X) + Var(Y) + 2\,Cov(X, Y)$.

(d) *(A portfolio.)* Two assets have random returns R_1, R_2, each with variance σ^2 and correlation ρ. Find the variance of the equal-weighted portfolio return $\frac{1}{2}R_1 + \frac{1}{2}R_2$. What happens as $\rho \to 1$? As $\rho \to -1$? In one sentence: why is this the entire logic of diversification?

(e) Define the prediction error $u \equiv y - \mathbb{E}[y|x]$. Show that for any function $g(x)$, $\mathbb{E}[g(x)\,u] = 0$. (Hint: law of iterated expectations.) In one sentence: why does this result mean that $\mathbb{E}[y|x]$ is the "best" predictor of y given x?

2 Conditional expectations and total variance

Suppose (y, x) are bivariate normal with

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \qquad \Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}$$

so $Var(y) = 4$, $Var(x) = 1$, and $Cov(y, x) = 1$. Recall that for the bivariate normal, $\mathbb{E}[y|x]$ is exactly linear: $\mathbb{E}[y|x] = \alpha + \beta x$.

(a) Solve for α and β using only the operator rules. (Hint: take expectations of both sides for one equation; take the covariance of both sides with x for the other.)

(b) For the bivariate normal, $Var(y|x)$ does not depend on x. Use the law of total variance, $Var(y) = \mathbb{E}[Var(y|x)] + Var(\mathbb{E}[y|x])$, to solve for $Var(y|x)$.

(c) Show that $\frac{Var(\mathbb{E}[y|x])}{Var(y)} = \rho^2$, where ρ is the correlation between y and x. (This is the population R^2 of the regression of y on x.)

(d) Now simulate 10,000 draws from this distribution. Verify (a)–(c) numerically: regress y on x and compare the coefficients to (a); compare the residual variance to (b); compare the R^2 to (c). Plot the mean of y within bins of x against the fitted line.

3 Watching the theorems work

Let $X_i \sim$ Exponential(1) (so $\mu = 1$ and $\sigma^2 = 1$, and the distribution is highly skewed).

(a) In Lecture 1 we derived $Var\left(\bar{X}_n\right) = \sigma^2/n$ from the operator rules alone. For each $n \in \{10, 100, 1000\}$, simulate $S = 1{,}000$ sample means $\bar{X}_n^{(s)}$ and compare the variance of the simulated sample means to σ^2/n. Present the comparison in a small table.

(b) Plot histograms of $\bar{X}_n^{(s)}$ for each n. Which theorem are you watching, and what is it doing to the center and spread?

(c) Now plot histograms of the *standardized* means $\sqrt{n}\left(\bar{X}_n^{(s)} - 1\right)$ for each n, overlaying a $\mathcal{N}(0, 1)$ density. Which theorem are you watching now? At what n does the normal approximation start looking trustworthy for this (skewed) distribution?

4 Omitted variables, with the answer key in hand

Recall the "silly example" from Lecture 2: rectangular paintings with $\ln A_i = \ln W_i + \ln H_i + \varepsilon_i$, so the true coefficients are $(1, 1)$ by geometry.

(a) Simulate 10,000 paintings with

$$\begin{pmatrix} \ln W_i \\ \ln H_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{wh} \\ \sigma_{wh} & 1 \end{pmatrix}\right), \qquad \varepsilon_i \sim \mathcal{N}(0, 0.01)$$

for $\sigma_{wh} = 0.6$. Run the *long* regression of $\ln A$ on $\ln W$ and $\ln H$, and the *short* regression of $\ln A$ on $\ln W$ only. Report both.

(b) Using the omitted variables bias formula from Lecture 2, compute what the short regression coefficient *should* be, and compare to (a).

(c) Repeat (a) for $\sigma_{wh} \in \{-0.6, 0, 0.3, 0.9\}$ and plot the short-regression coefficient against σ_{wh}, with the OVB-formula prediction overlaid as a line. When is the short regression fine?

(d) Two sentences: your colleague says "I get a precise, highly significant coefficient of 1.6; the standard error is tiny, so I trust it." What did they get wrong, and why didn't the standard error warn them?